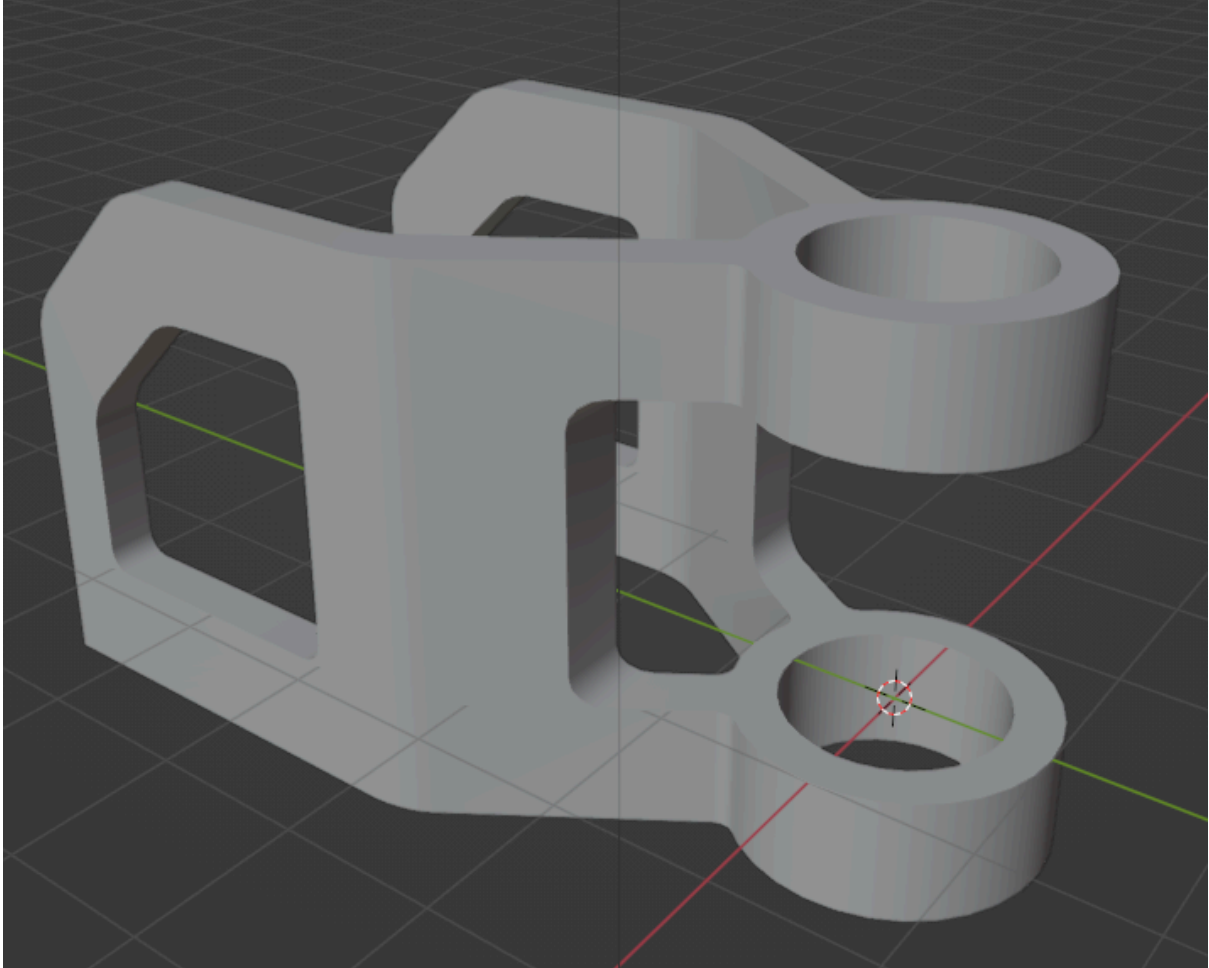
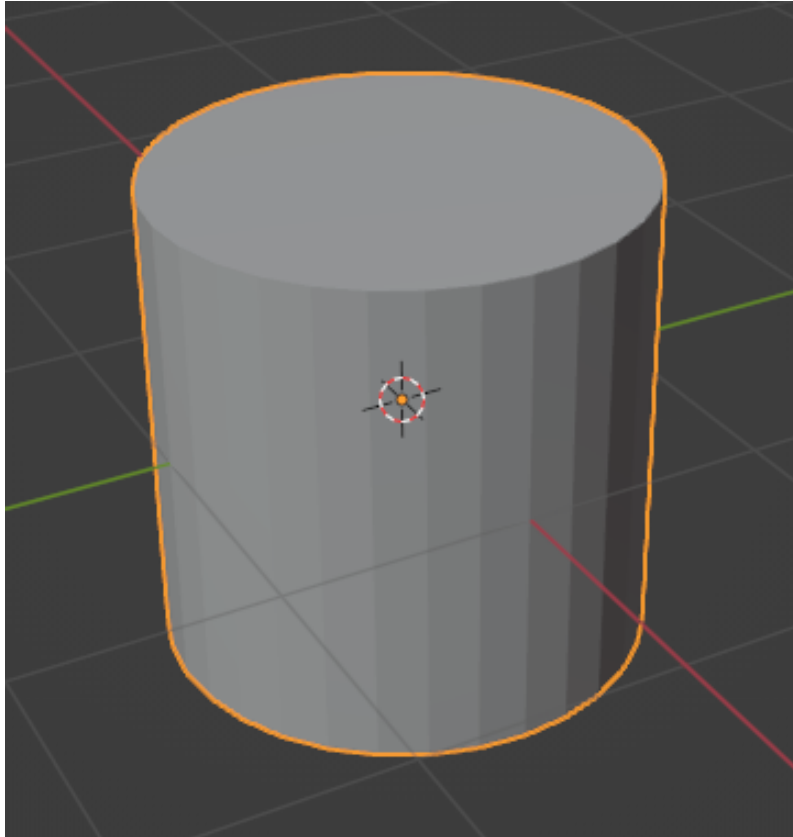


## Hard Surface Model 2

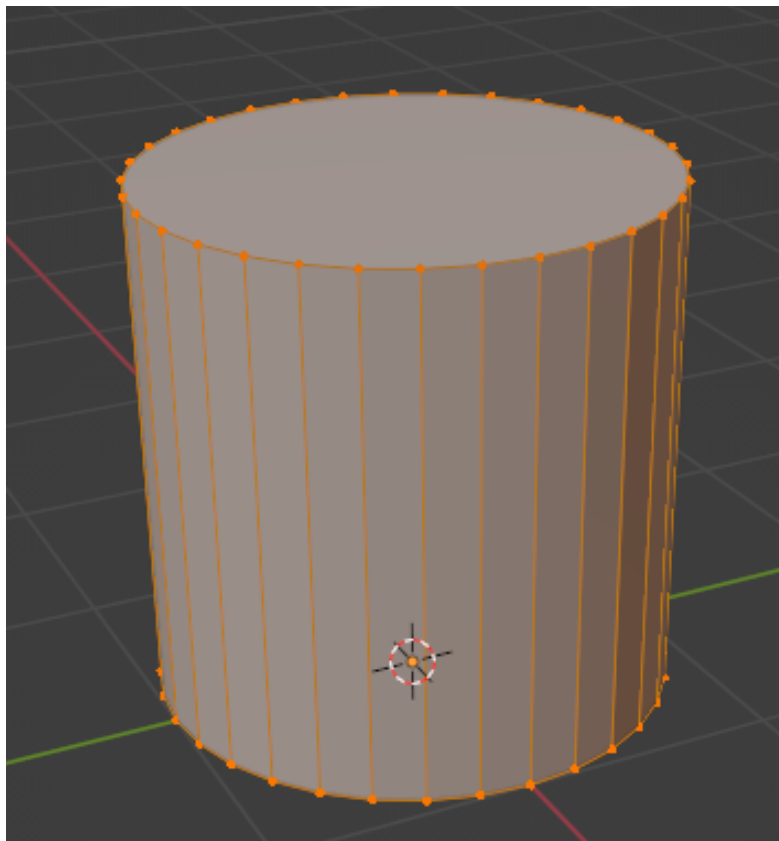
Important concepts covered: boolean modifier, change viewport display, knife tool



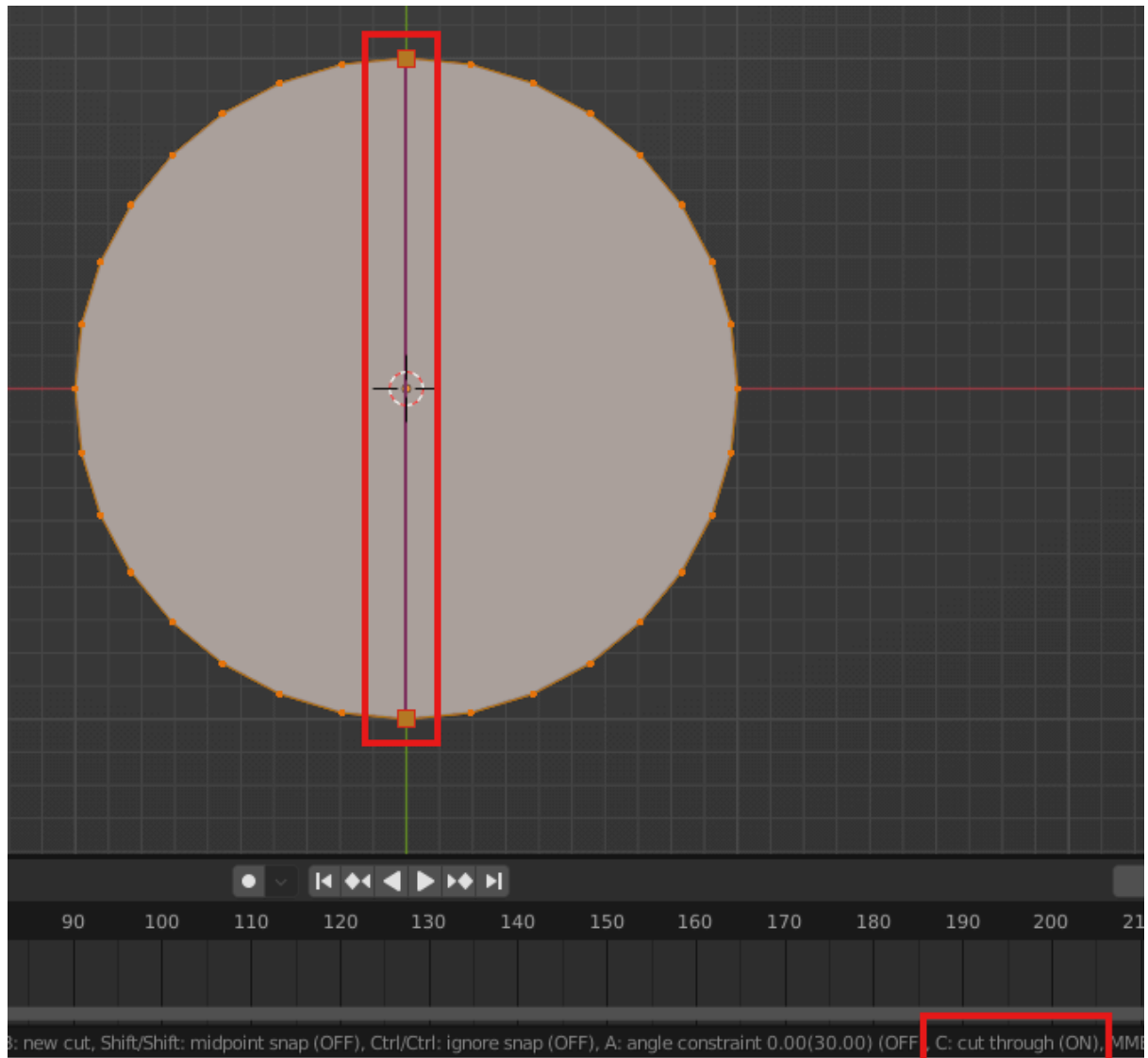
First, delete the default cube (DEL or X) and add a cylinder (SHIFT+A → MESH → Cylinder):



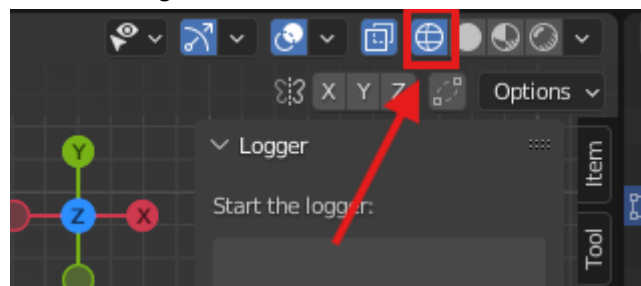
Enter edit mode and move it along Z axis (G+Z) by 1:

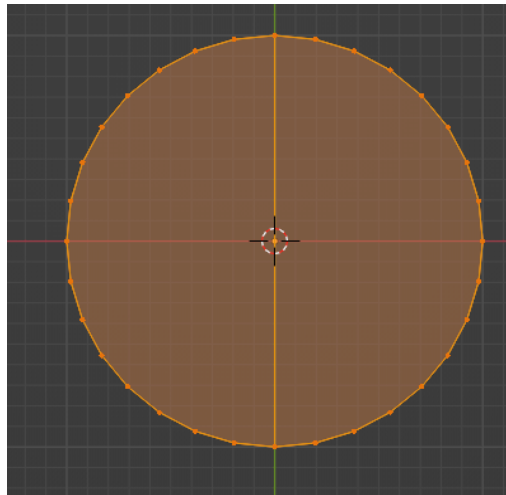


Go to the top view (NUMPAD 7), press “K” to use the knife tool and then press “C” to enable “Cut Through” and cut the mesh along the Y axis (click in the bottom vertex, then in the top vertex and click Enter):

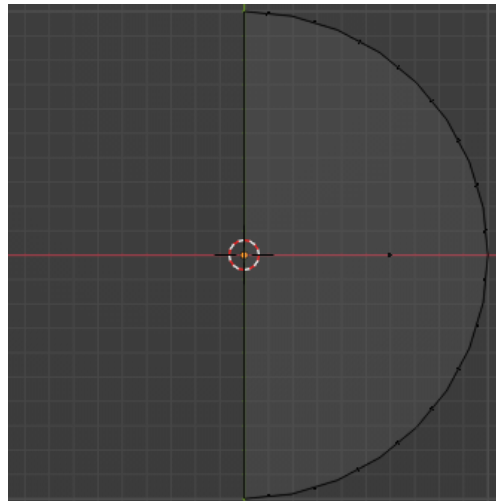


Now change to wireframe shading view:

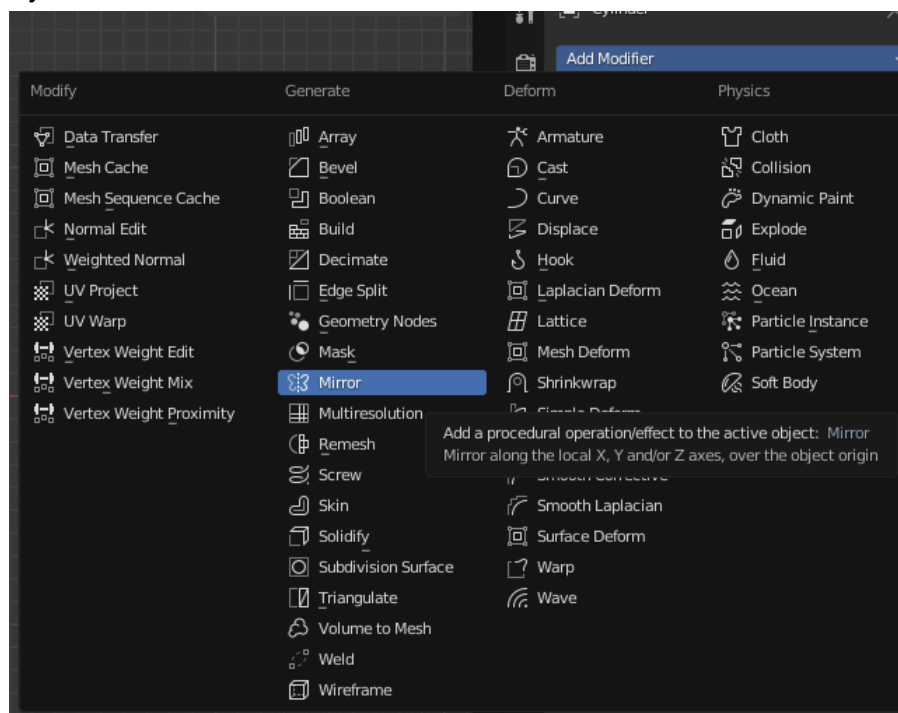




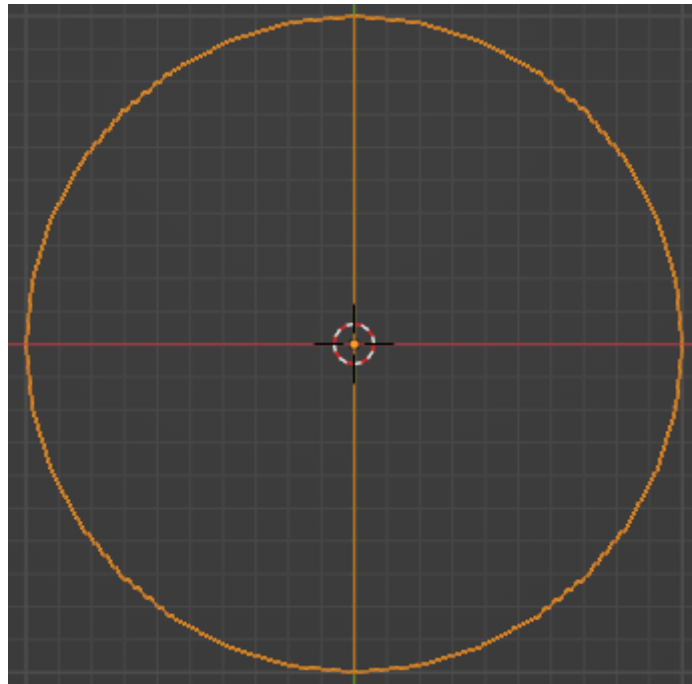
Change to face selection mode, click and drag to select all left part of the mesh and delete it (DEL or X):



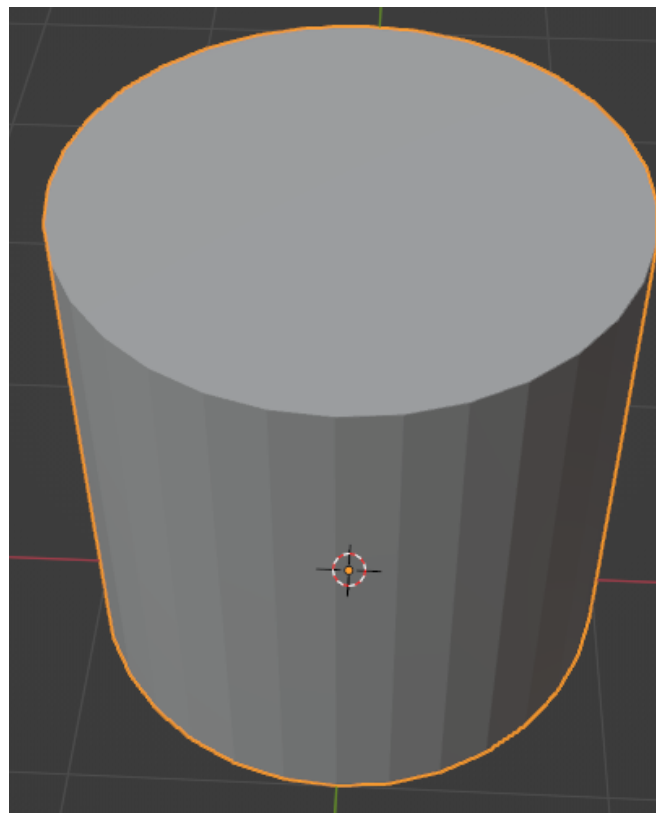
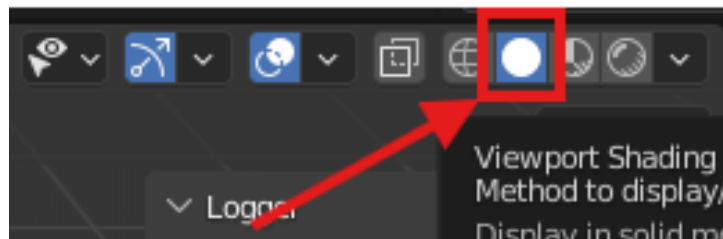
Now go to object mode and add a mirror modifier:



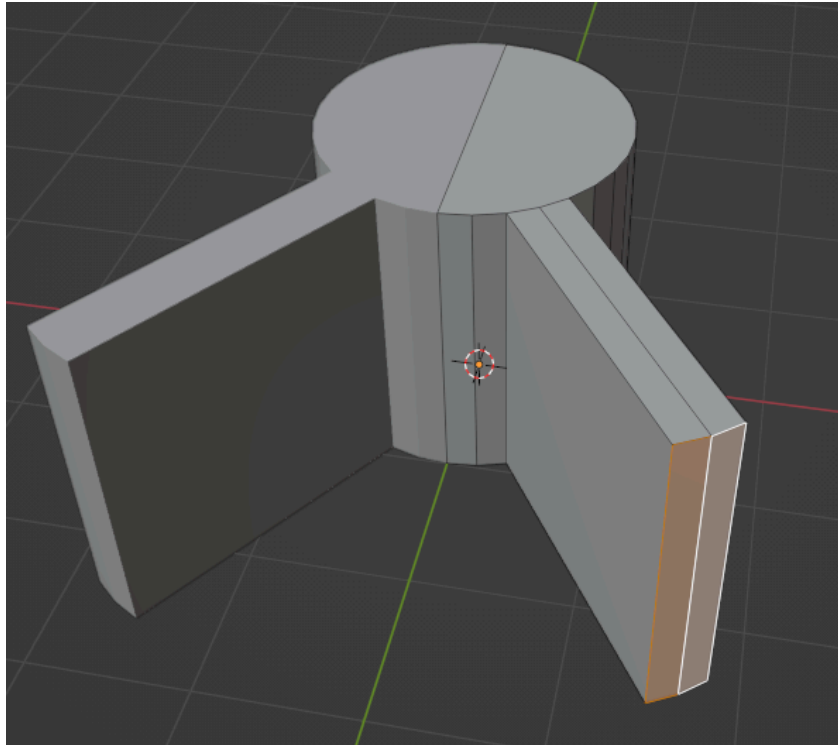
The cylinder should now be complete again:



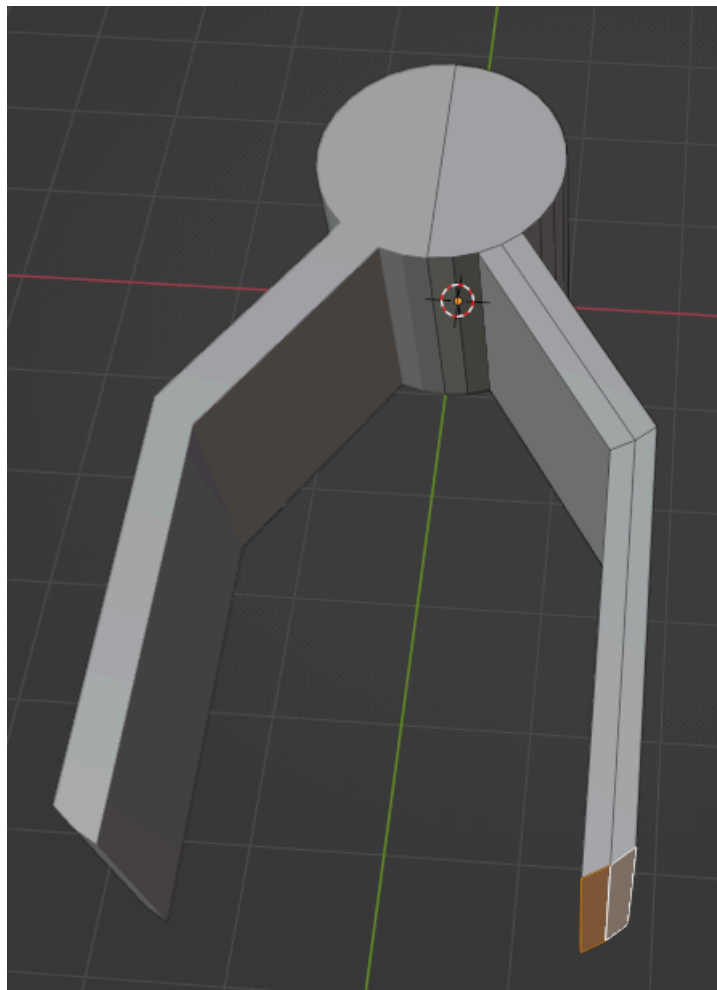
Change back to solid shading view again:



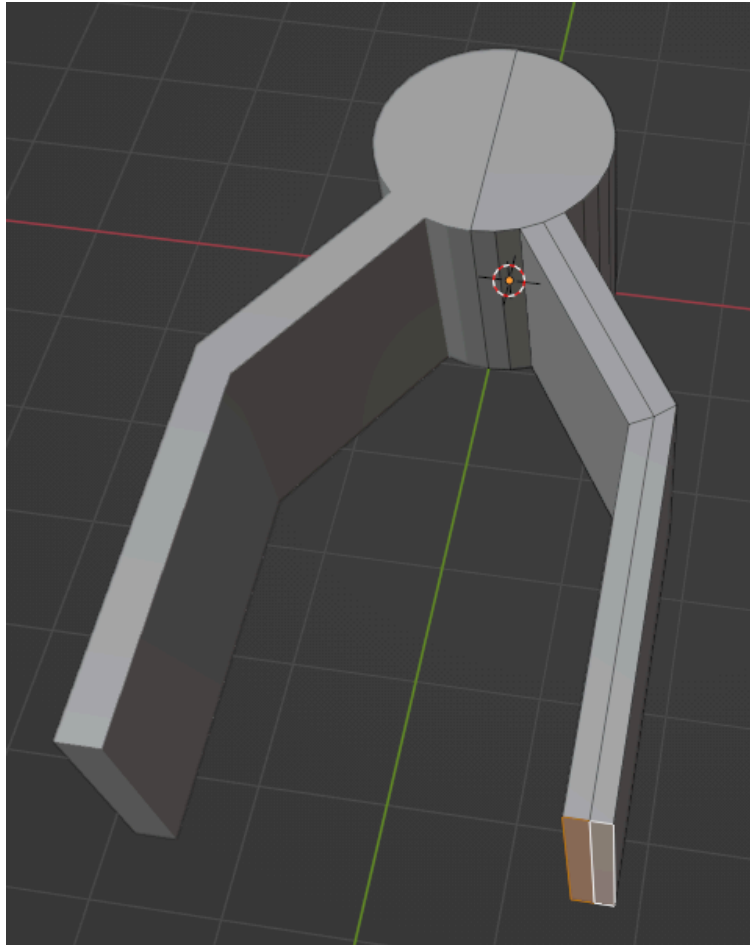
Change to edit mode again select the following faces and extrude (E) them by 2 units:



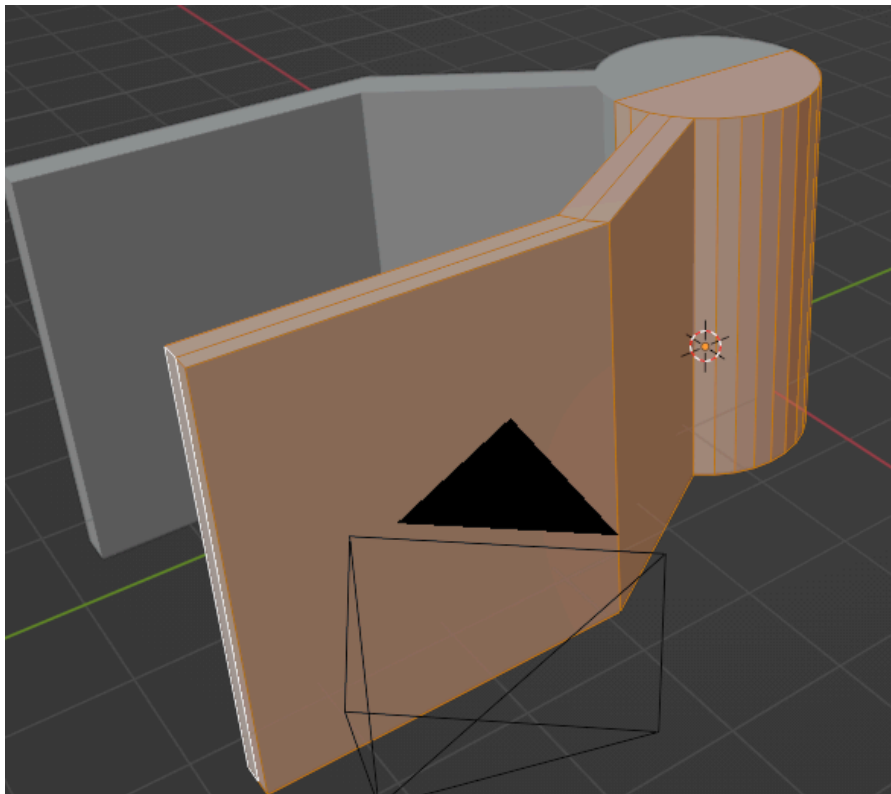
Now extrude again but now along the Y axis (E+Y) by -3:



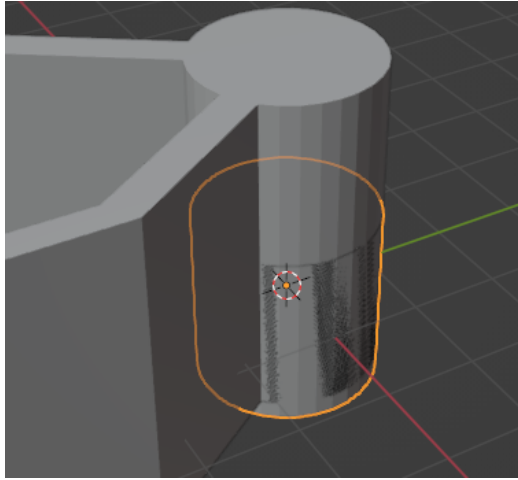
Next, still with the same faces selected, scale it along the Y axis (S+Y) by 0:



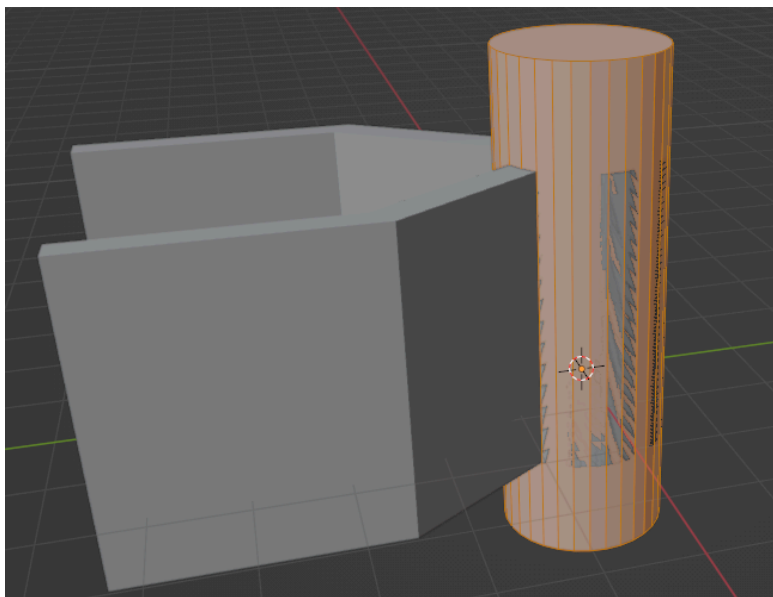
Now select all by pressing "A" and scale along the Z axis (S+Z) by 1.8:



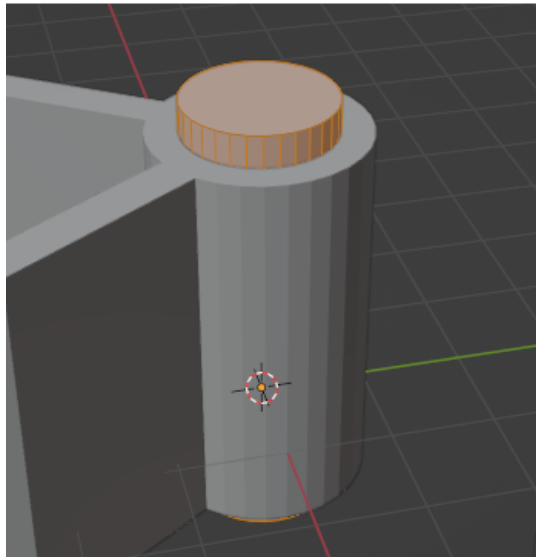
Now enter object mode, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a new cylinder object (SHIFT+A → MESH → Cylinder) into the scene:



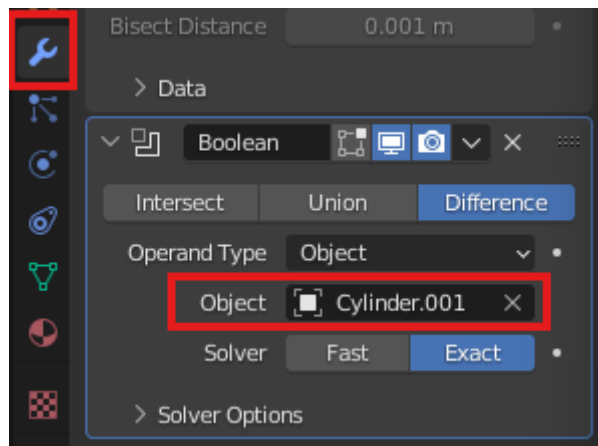
Enter edit mode on this new object, scale along the Z axis (S+Z) by 3 and move it along the Z axis (G+Z) by 1:



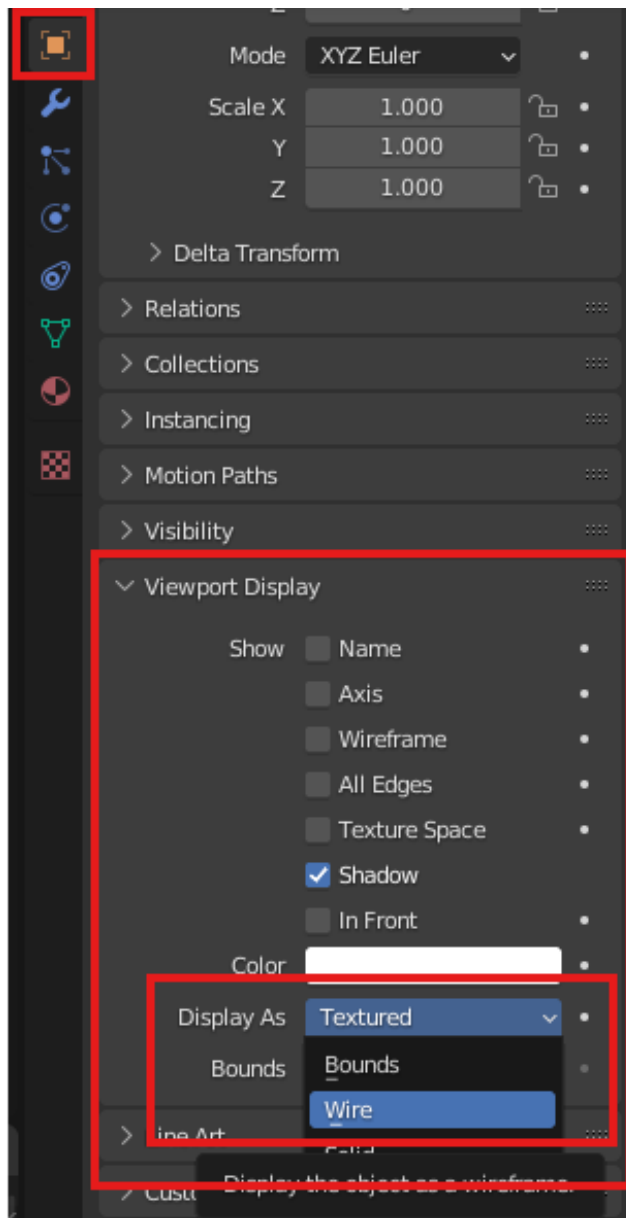
Now scale it (S) by 0.7:



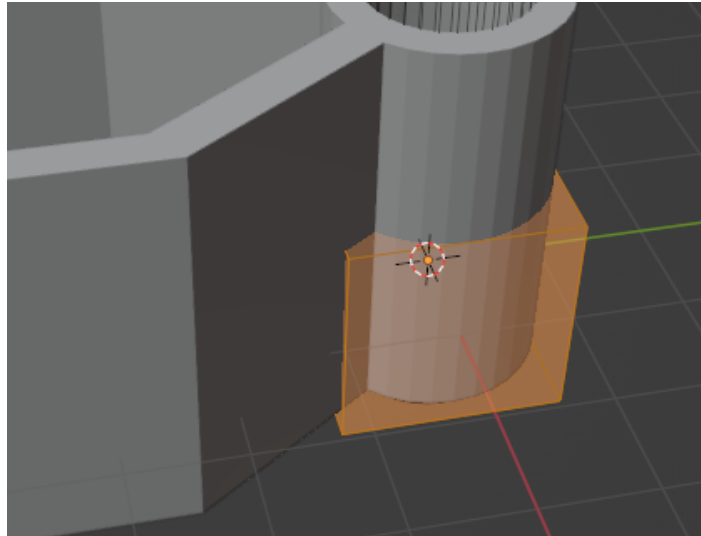
Now enter object mode, select the other object and add a boolean modifier. Under “Object”, select Cylinder.001:



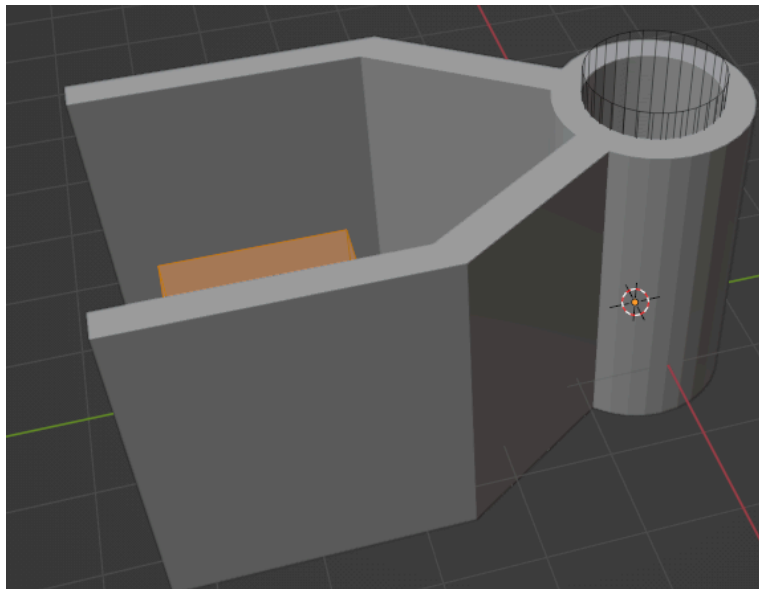
Now select the cylinder again, go to “Object Properties” → “Viewport Display” → “Display As” and select “Wire”:



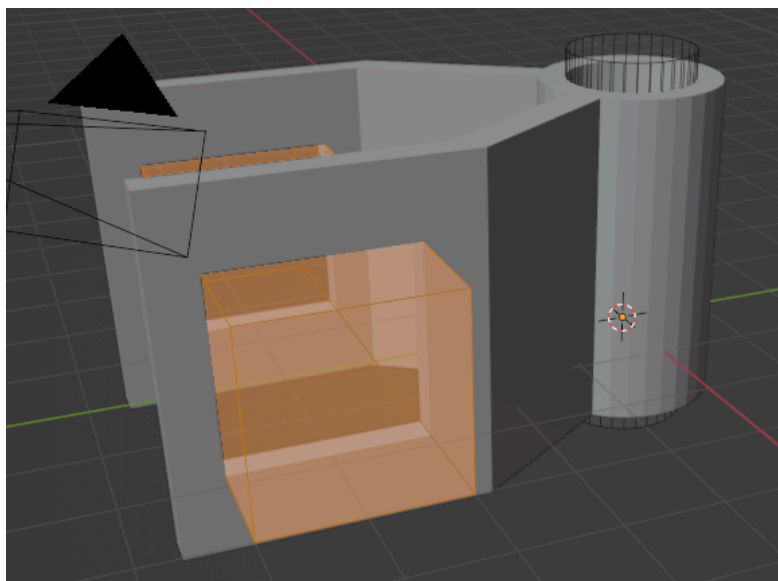
Now enter edit mode on the cylinder, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a cube mesh (SHIFT+A → MESH → Cube):



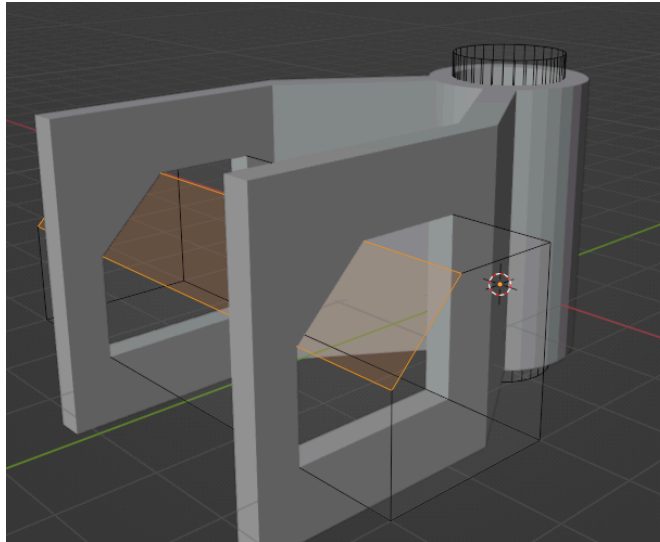
Move the cube along the Y axis (G+Y) by -4:



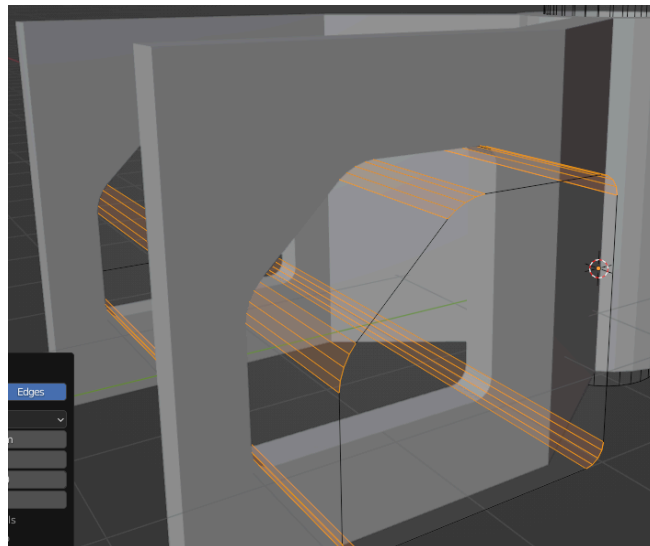
Now scale it along the X axis (S+X) by 3 and move it along the Z axis (G+Z) by 1:



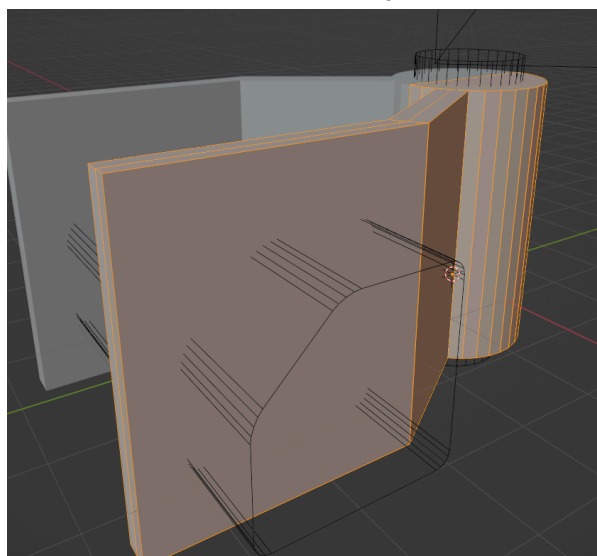
Change to Edge selection mode and bevel (CTRL + B) the following edge by 0.8:



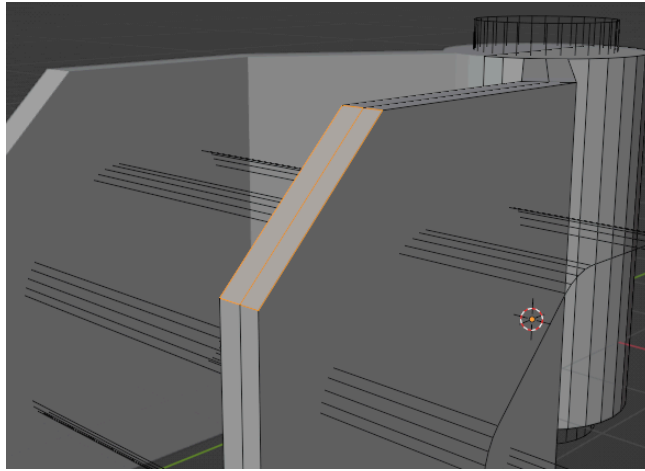
Now add another bevel (CTRL+B) to all the X-parallel edges by 0.15 and 4 segments (use the scroll wheel of the mouse to increase the number of segments):



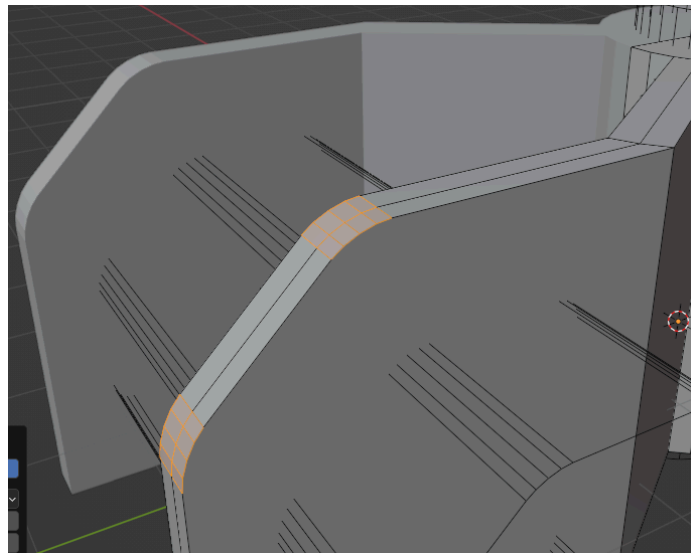
Now go back to Object mode and select the other object and enter edit mode:



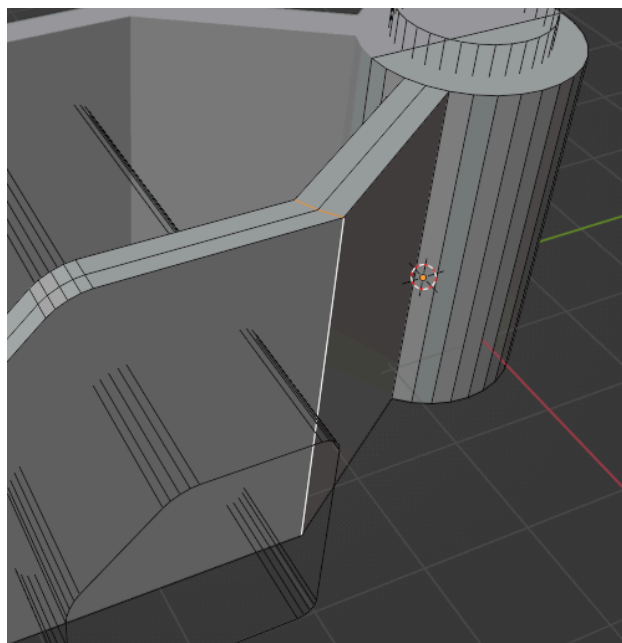
Bevel (CTRL+B) the top left corner horizontal edges by 1:



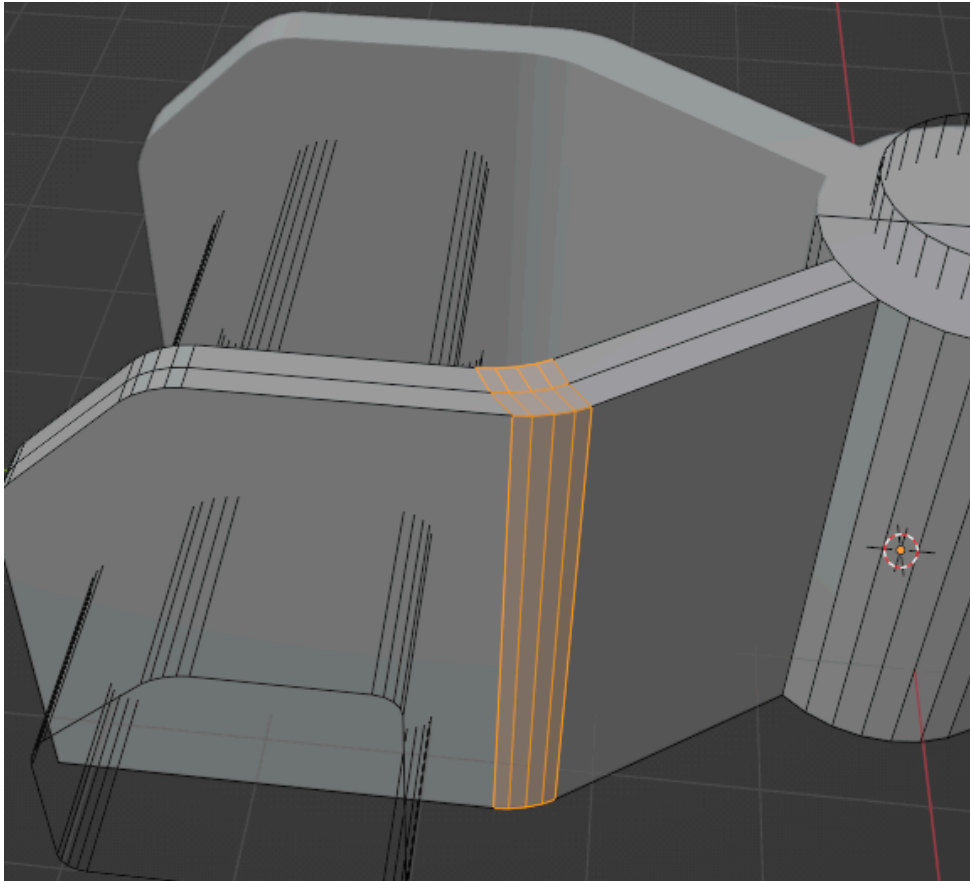
Now select the horizontal edges created in the “ramp” and bevel (CTRL+B) them by 0.2 and 4 segments:



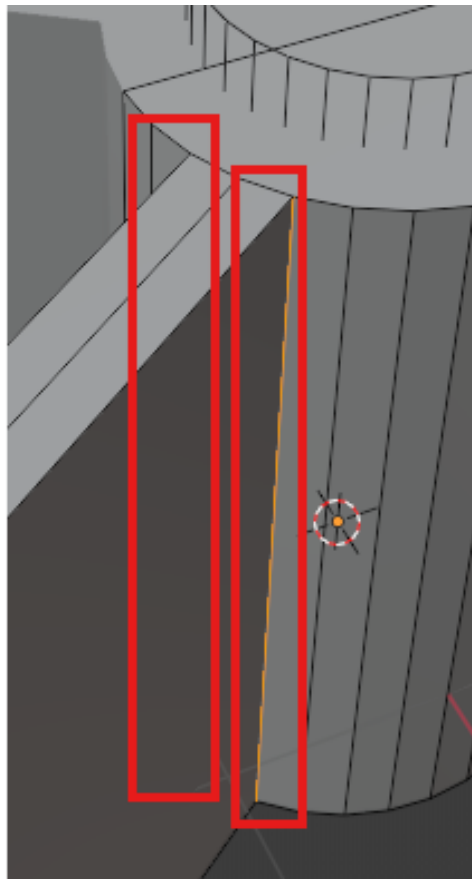
Next, select the following edge loop (ALT + left click one of the edges of the loop):



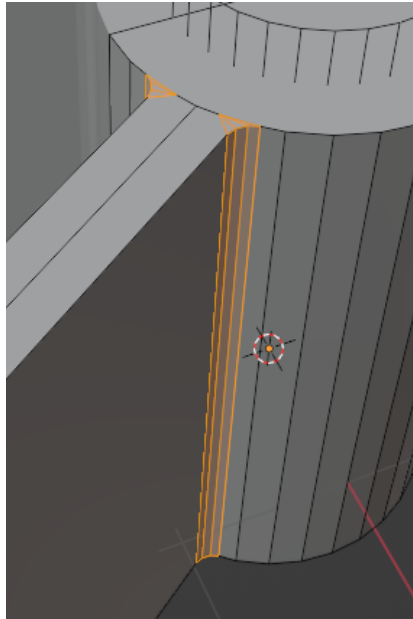
And bevel (CTRL+B) them by 0.2 and 4 segments again:



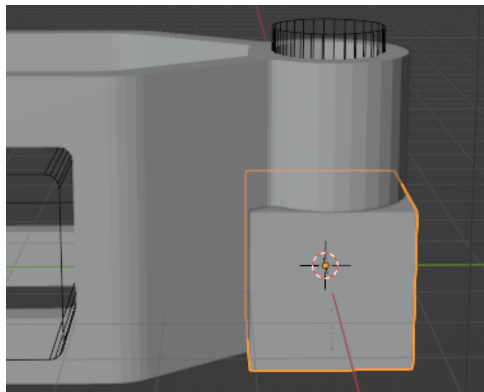
Now select the 2 vertical edges connecting to the cylinder:



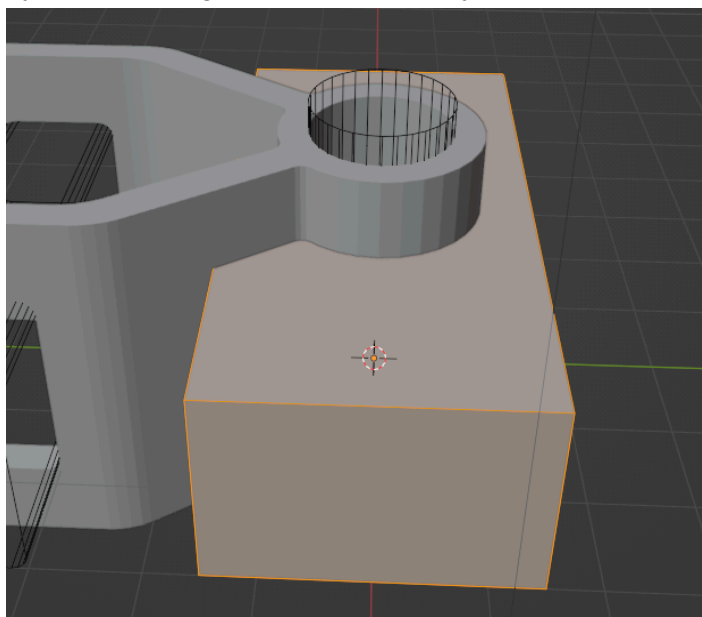
And bevel (CTRL+B) them by 0.09 and 3 segments:



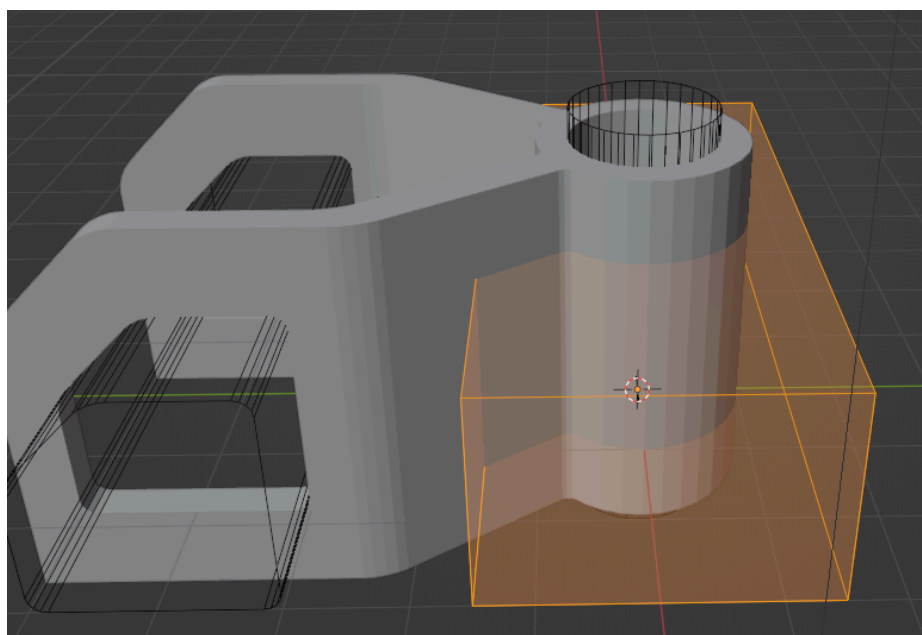
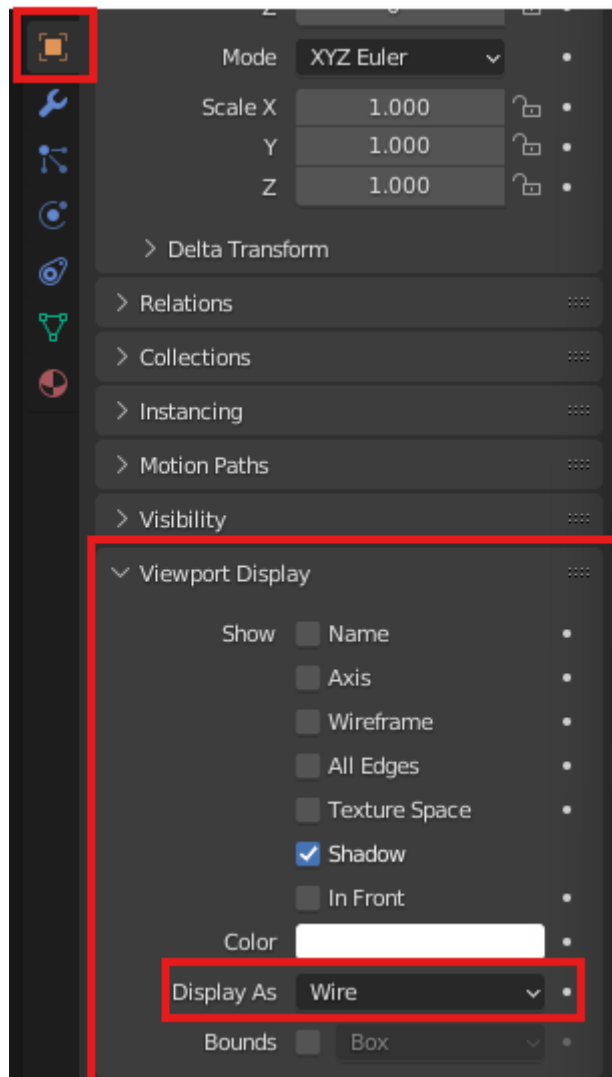
Now go back to object mode, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a cube object to the scene (SHIFT+A → MESH → Cube):



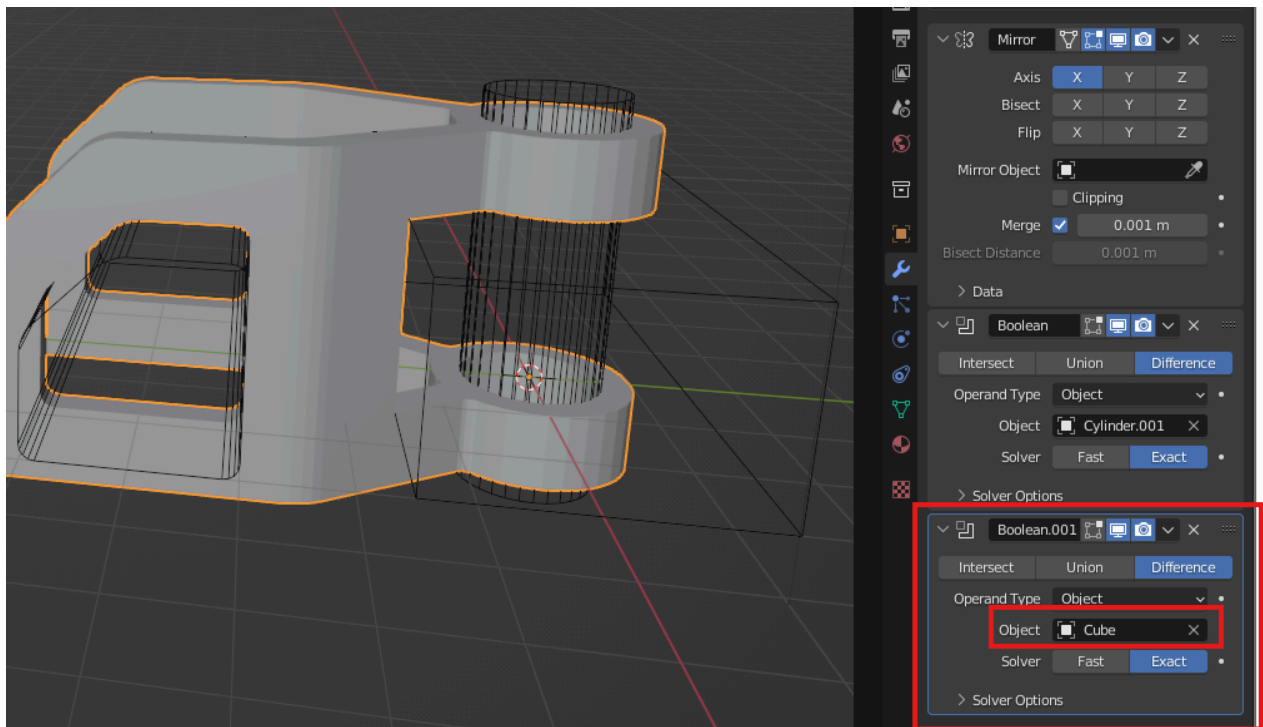
enter edit mode and move the mesh along the Z axis (G+Z) by 1, scale it along the X axis (S+X) by 3 and finally scale it along the Y axis (S+Y) by 1.5:



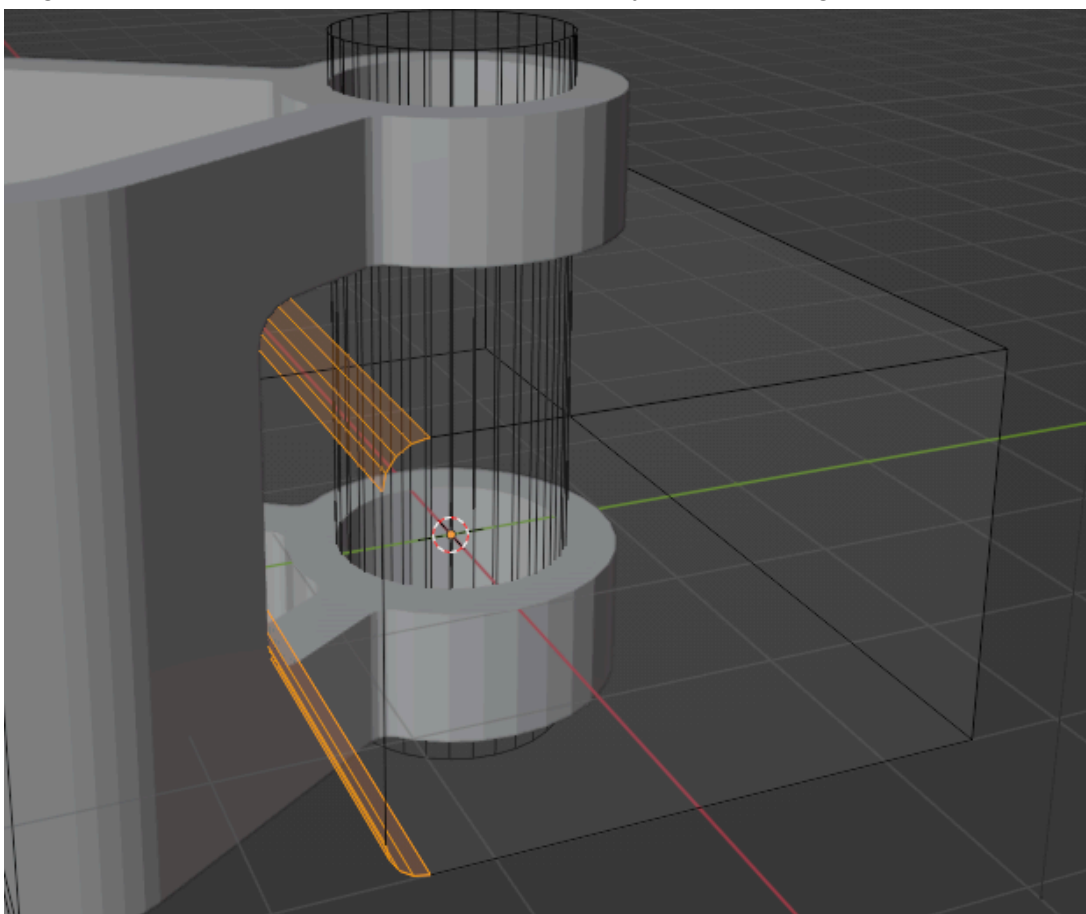
Now change to display it as “Wire” as we did before by going to “Object Properties” → “Viewport Display” → “Display As” and select “Wire”:



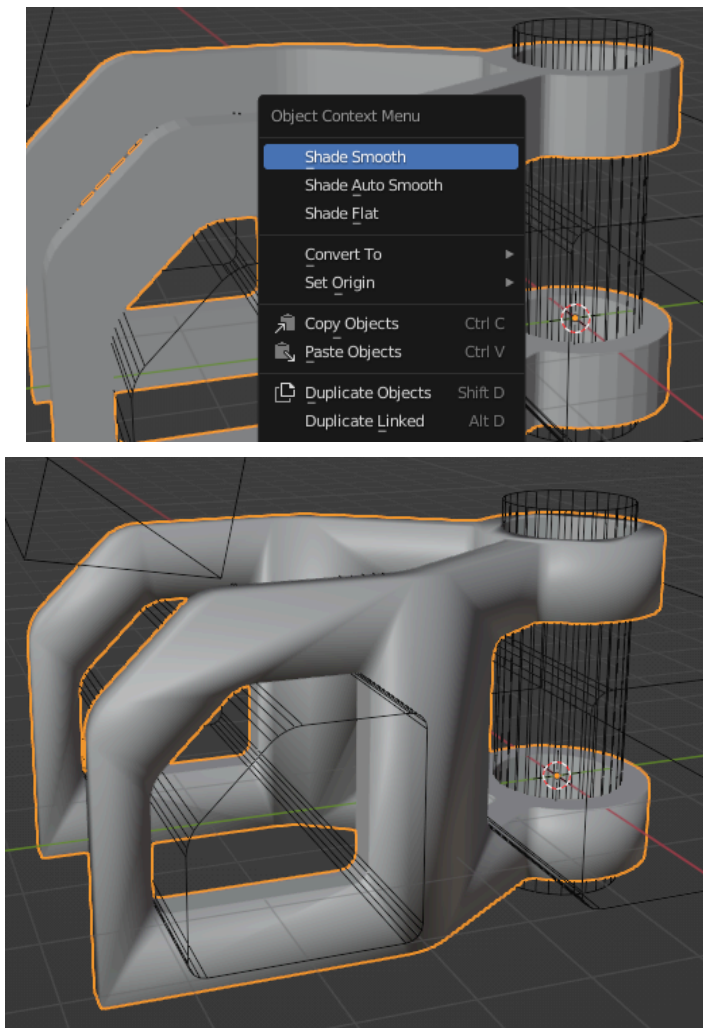
Now go back to Object mode, select the main object (the one with the holes) and add another boolean modifier and select “Cube” as the Object:



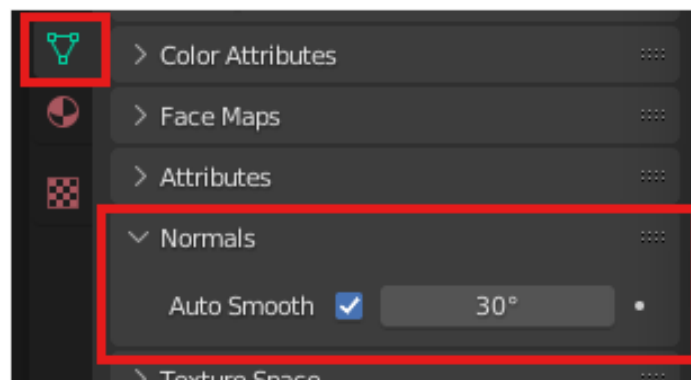
Now select the created cube again and enter edit mode. Select the two X-parallel edges touching the main mesh and bevel (CTRL+B) them by 0.2 and 4 segments:

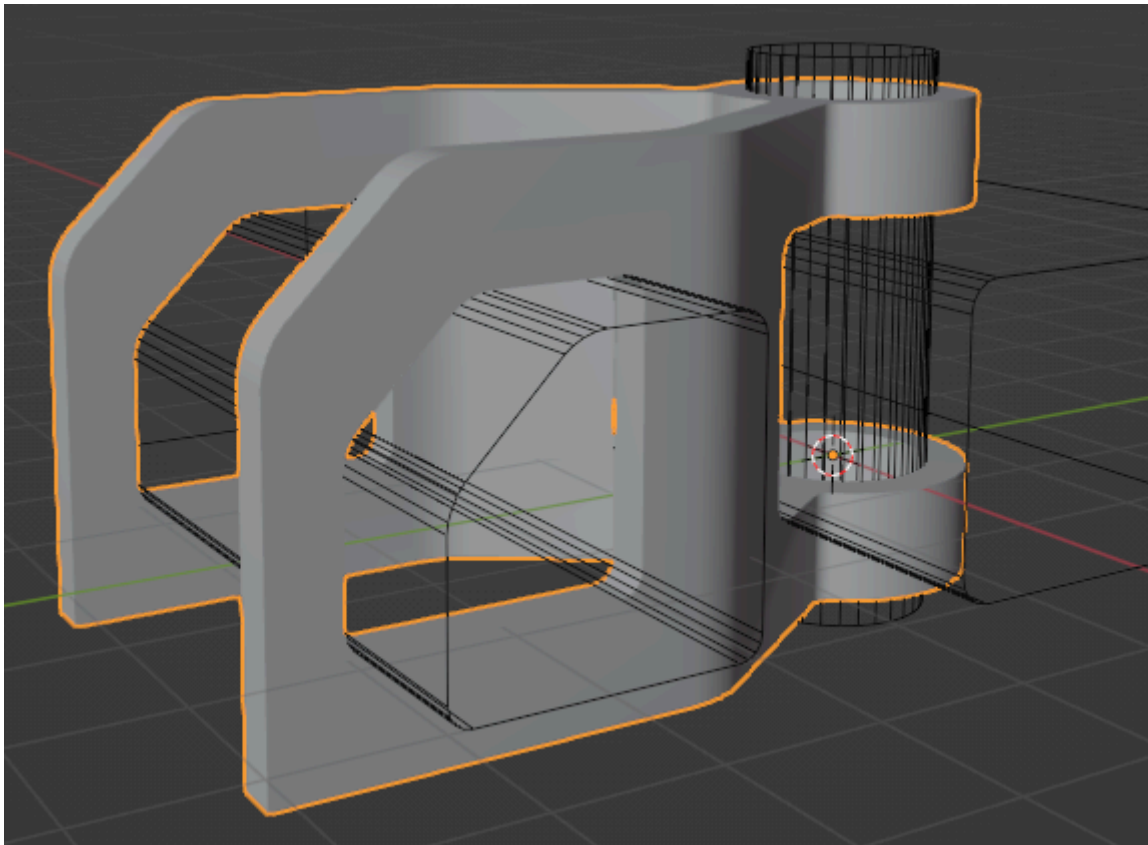


Now go to object mode again and select the main object. Right click the viewport and select “Shade Smooth”:

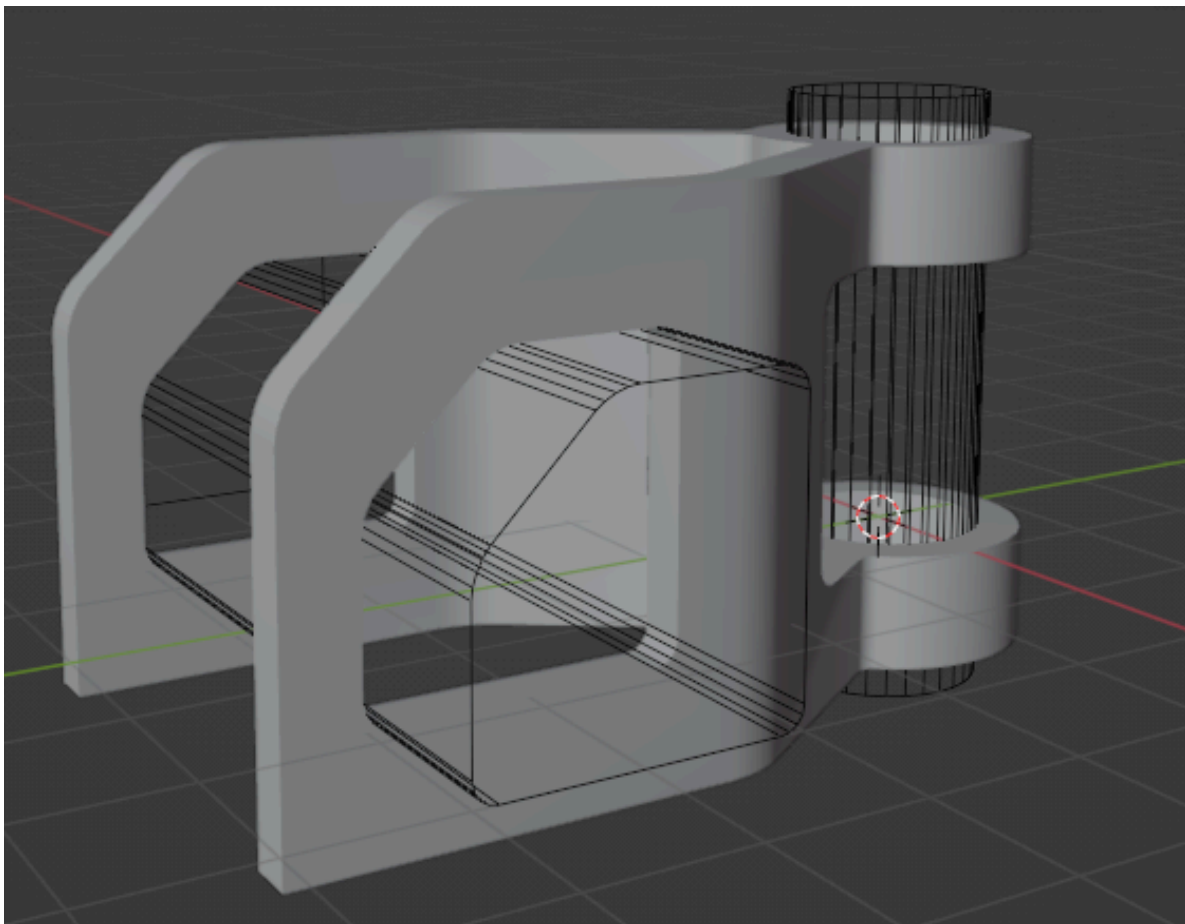


As you can see the mesh looks “weird”. That’s because the smoothness is being applied to all vertices. But we just want it to be applied in the “rounded” edges, not the perpendicular ones. So in order to fix it, go under “Object Data Properties” → “Normals” and check “Auto Smooth”:

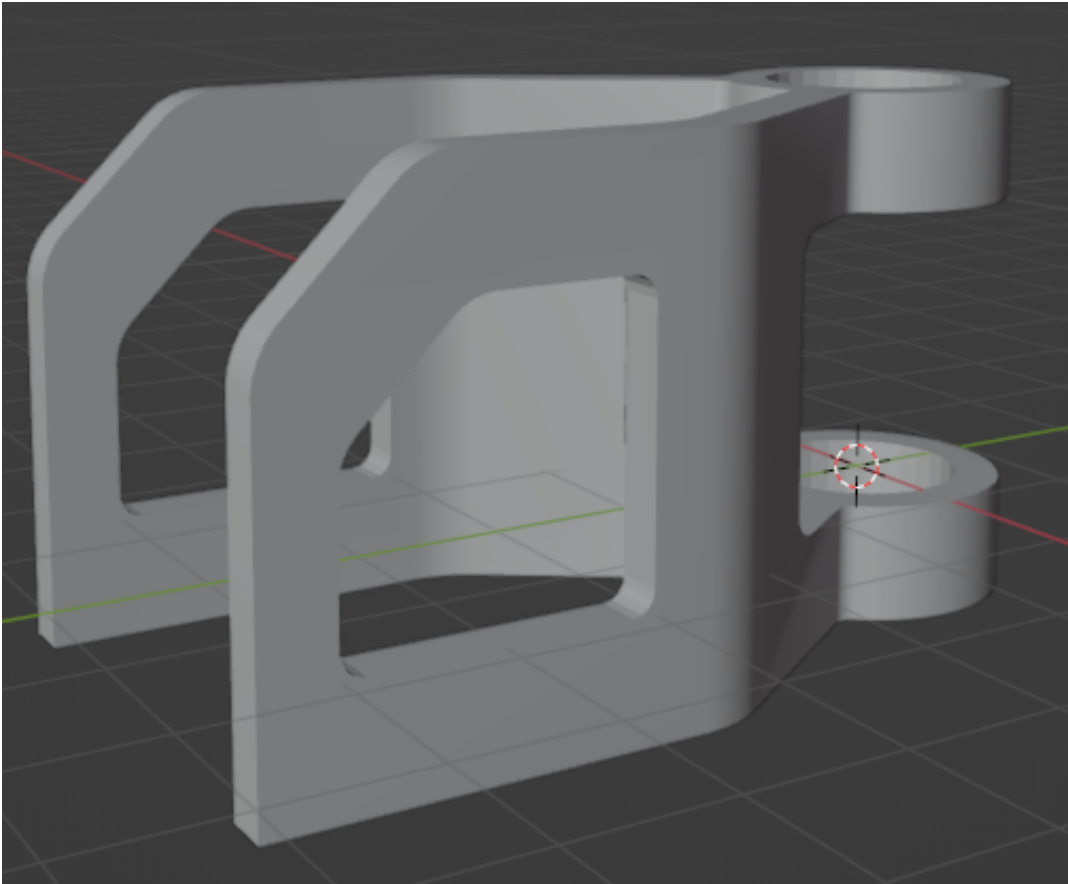




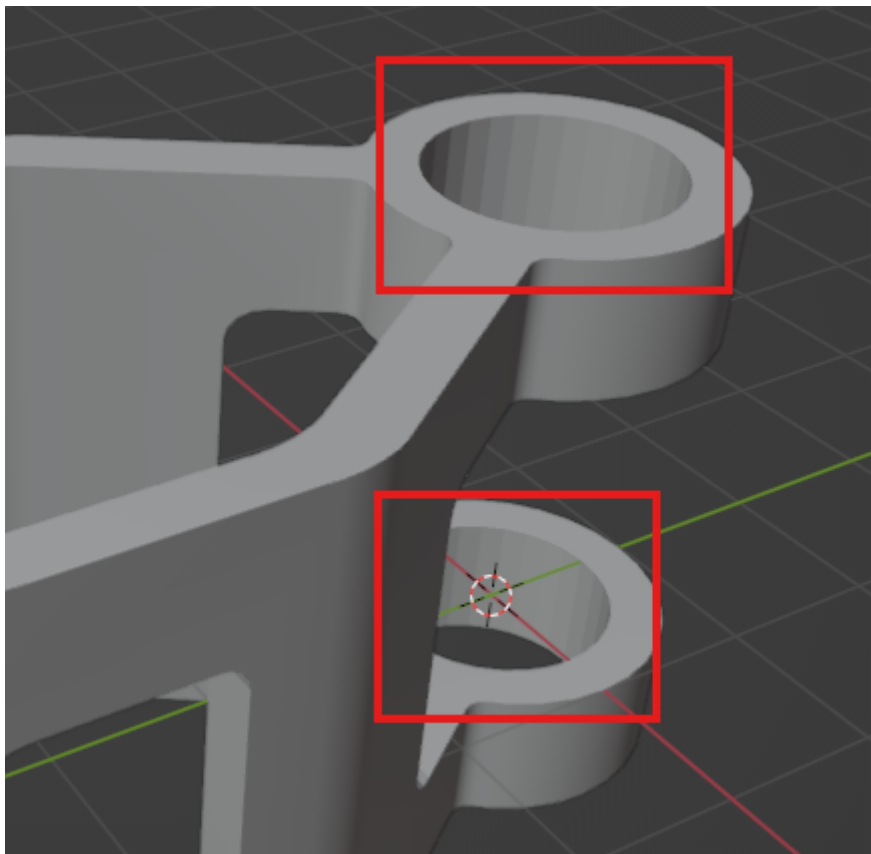
Now select the Cube and hide it by pressing “H”:



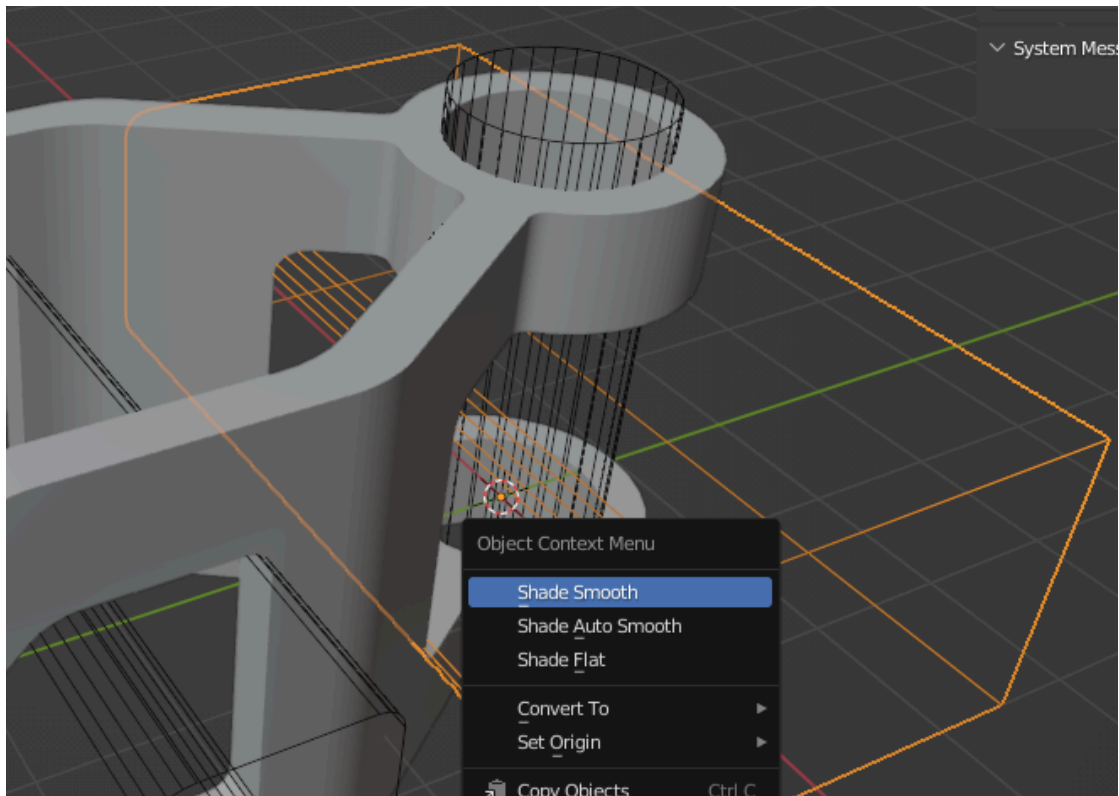
Repeat this also to the other object cutting through the main one:



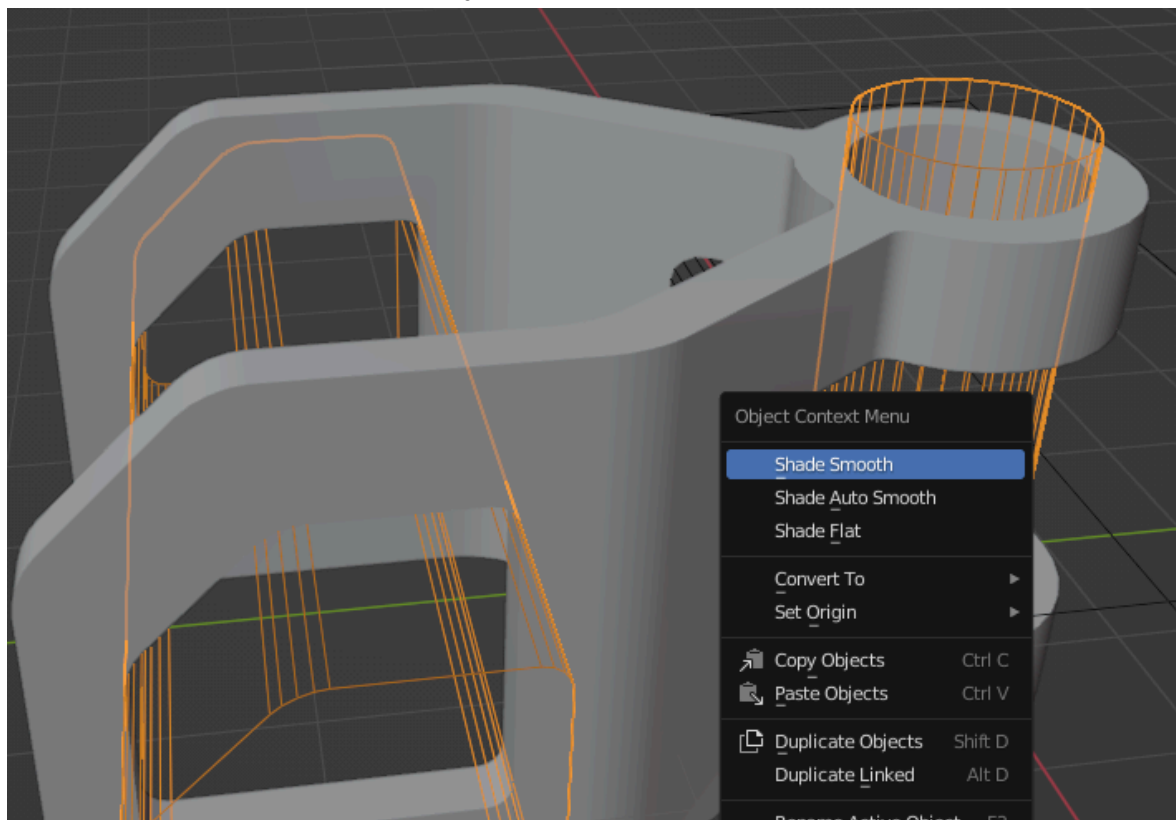
As you can see, although the mesh seems smoothed outside, it is not inside:



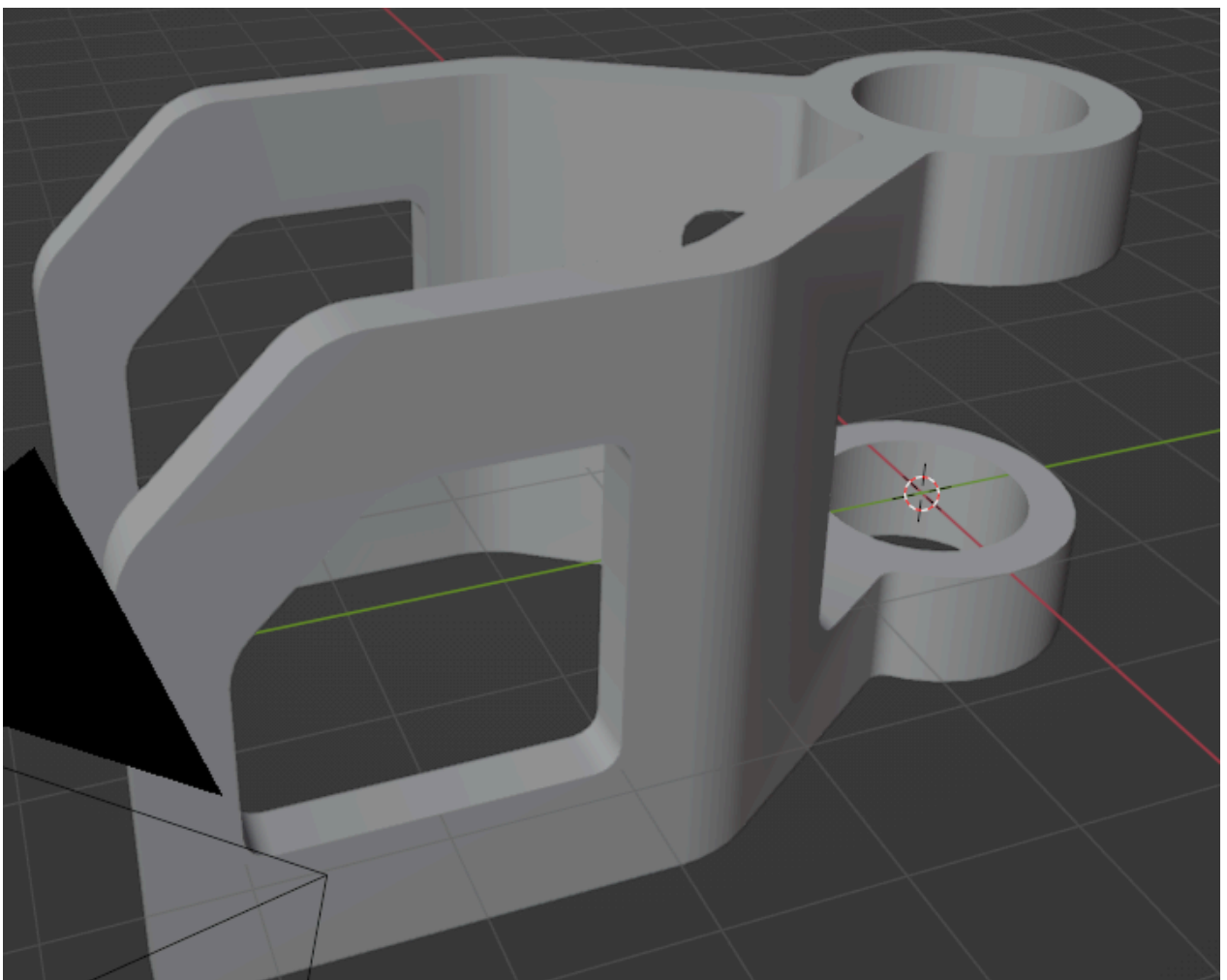
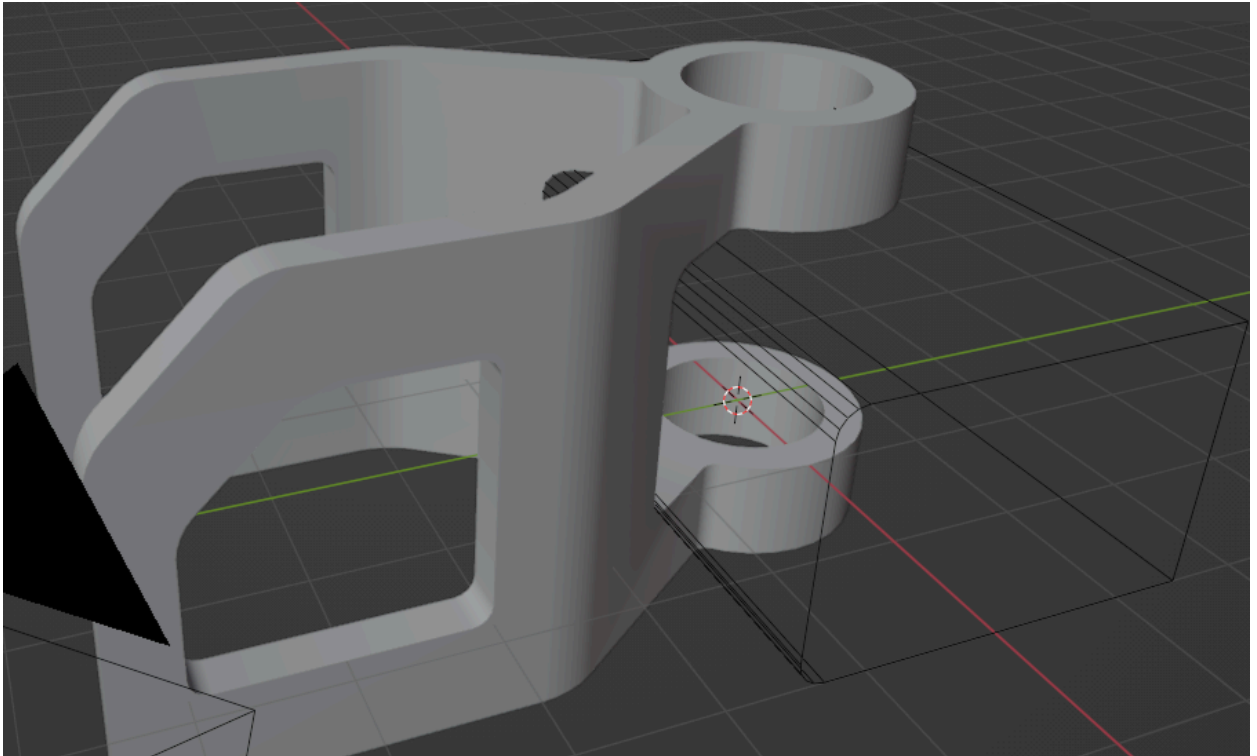
This is because the objects that are cutting the mesh are not smoothed. So let's unhide the objects by pressing ALT + H. Select the cube, right click the viewport and select "Shade Smooth":



Do the same to the other boolean object:



Now hide this object and then hide the cube again:



Finally, you now have modeled an industrial piece!

